

CHAPTER TWO

DEFINING RESEARCH PROBLEM AND HYPOTHESIS FORMULATION

What is a research problem?

- ✦ All research is set in motion by the existence of a problem
- A problem might be defined as the **issue that exists in the literature, theory, or practice** that leads to a need for the study.
- ✦ The problem identification and explanation affect the **quality, usefulness, effectiveness, and efficiency** of the research, more than any other part of the research plan.

Cont'd

- The prospective researcher should think on **what caused the need to do the research** (problem identification).

The question that he/she should ask him/herself is:
Are there questions about this problem to which answers have not been found up to the present?

- we have the problem only **if we do not know what course of action is best,**
- and must be in doubt about the solution.

- A problem is a **perceived difficulty**, a feeling of **discomfort** about the way things are or a **discrepancy between what some one believes should be** the situation and **what the situation is in reality**.
- While problems are initiating forces behind research, not all problems require research.
- a researcher must examine all the symptoms (presented to him or observed by him) concerning a problem before he can diagnose correctly.

Cont'd

A potential research situation arises when **three** conditions exist

- A **perceived discrepancy** exists between what is & what should be.
- A **question** exists about **why** there is a discrepancy.
- At least **two possible & plausible answers** exist to the question.

“Problem clearly stated is half solved”

the components of a research problem

- (i) There must be an individual or a group which has some difficulty or the problem.
- (ii) There must be some objective(s) to be attained at.
- (iii) There must be alternative means. This means that there must be *at least two means* available to a researcher.
- (iv) There must remain some doubt in the mind of a researcher with regard to the selection of alternatives.

Why is it important to state & define the problem?

- ✦ Is the **foundation** for further development of the **research proposal** (objectives, methodology, work plan, budget, etc.)
- ✦ **Makes it easier** to find information & reports of **similar studies** from which your own study design can benefit.
- ✦ Enables you to systematically point out why the **proposed research on the problem should be undertaken & what you hope to achieve with the study results.**

selecting a research problem

- i. **Subject which is overdone** should not be normally chosen
- ii. **Provocative subject** should not become the choice of an average researcher.
- iii. Too narrow or **too vague problems** should be avoided.
- iv. The **subject selected** for research should **be familiar** and **feasible**.

V. The **importance of the subject**, the qualifications and the training of a researcher, the costs involved, the time factor must also be considered.

In other words, before the final selection is made a researcher must ask himself the following questions:

- (a) **Whether you are well equipped in terms of your background to carry out the research?**
- (b) Whether the study **falls within the budget** he can afford?
- (c) Whether the **necessary cooperation can be obtained** from those who must participate in research as subjects?

Vi. The selection of a problem must be preceded by a preliminary study.

- This may not be necessary when the problem requires the conduct of a research closely similar to one that has already been done.

How to write a problem statement?

- Problem statement should be
 - **Precise & concise**, while not forgetting to mention essential points.
- *Example:* Ambiguous definition: “Find out how much, sales have declined recently”. Let us suppose that, the research problem is defined in broad and general way as follows:
- “Why is the productivity in **Korea** is very much higher than in **India**”?
- In this type of question, a number of ambiguities are there, such as:
 1. What sort of productivity is to be specified; Is it men, machine, materials? Etc.
 2. To which type of industry, the productivity is related to?
 3. What period of time, the productivity is being talked about?

What information should be included in the problem statement?

- Nature of the problem
- Major factors that may influence the problem
- Justification for the study
- Type of information expected

Cont. . . .

■ *NATURE OF THE PROBLEM*

- 🖱️ **Description** of the discrepancy **between what is & what should be.**
- 🖱️ Description of **the size, distribution & severity** of the problem
 - ✳️ **who is affected, where, since when & what are the consequences for those affected.**
- 🖱️ For a descriptive or evaluation study the different components of the problem should be elaborated.

Cont'd

- ***FACTORS INFLUENCING THE PROBLEM***
 - An analysis of the major factors that may influence the problem &
 - if the problem is to be fully understood A discussion of why certain factors need more investigation.

Cont'd

■ ***JUSTIFICATION FOR THE STUDY***

- A brief description of any solutions to the ***problem that have been tried in the past***
- How well they have worked, and
- Why is the problem important and **further research is needed.**

 **Convince others about the importance of the problem**

identifying deficiencies in the past literature,

- Identify specifically the **deficiencies of other studies** (e.g. **methodological flaws**, **variables overlooked**).
- Write about **areas overlooked by past studies**, including topics, **special statistical treatments**, **significant implications**, and so forth.
- Discuss **how a proposed study will remedy these deficiencies** and **provide a unique contribution to the scholarly literature**

Cont'd

- ***TYPE OF INFORMATION EXPECTED***
 - A description of the **type of information expected to result** from the project and
 - How this information will be used to help solve the problem.

TECHNIQUES AND APPROACHES TO PROBLEM DEFINITION

- 1. Statement of the problem in a general way:** First of all the problem should be stated in a broad general way, keeping in view either some practical concern or some scientific or intellectual interest.
- 2. Understanding the nature of the problem:** The next step in defining the problem is to understand its origin and nature clearly. The best way of understanding the problem is to discuss it with those who first raised it in order to find out how the problem originally came about and with what objectives in view.

- 3. Surveying the available literature:** All available literature concerning the problem at hand must necessarily be surveyed and examined before a definition of the research problem is given.
- 4. Developing the ideas through discussions:** Discussion concerning a problem often produces useful information.
- 5. Rephrasing the research problem:** Finally, the researcher must sit to rephrase the research problem into a working proposition.

PROBLEM FORMULATION, RESEARCH QUESTION AND HYPOTHESIS

- The *research problem* typically takes the form of a concise question regarding the relationship between two or more variables.
- When articulating a research question, it is critically important to make sure that the question is specific enough to avoid confusion and to indicate clearly what is being studied.
- A vague research question often results in methodological confusion & does not clearly indicate what or who is being studied.

Example

- (1) What effect does **weather** have on **memory**?
- (2) Does **exercise** improve physical and mental health?
- (3) Does taking street drugs result in **criminal behavior**?

Hypothesis

- is simply an educated—and testable—guess about the answer to your research question.
- A hypothesis is often described as an attempt by the researcher to explain the phenomenon of interest.
- A hypothesis represents a declarative statement of the relations between two or more variables.
- hypotheses attempt to explain, predict, and explore the relationship between two or more variables.

Cont'd

- Hypotheses can take various forms, depending on the question being asked and the type of study being conducted
- A key feature of all hypotheses is that each must make a *prediction*.
- These predictions are then tested by gathering and analyzing data, and the hypotheses can either be supported or refuted.
- Hypothesis is optional.

Null and alternative hypothesis

- The *null hypothesis* always predicts that there will be no differences between the groups being studied._
- By contrast, the *alternate hypothesis* predicts that there will be a difference between the groups.

Example

Ho: there will be no difference in terms of blood pressure levels between the group that receives the medication.

H1:there will be a difference between the two groups with respect to blood pressure levels.

- Ho: the exercise group and the no-exercise group will not differ significantly on levels of cholesterol.
- H1:the two groups will differ significantly on cholesterol levels.

END